The Impact of Physical Fatigue on the Gait Metrics of Speed, Power, Balance and Symmetry as Measured by SportGait BioKinetoGraph (BKG)

Michelle Messer

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Approved by

Advisory Committee

_____
Dr. Len Lecci *(Chair)*

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Dr. Rider

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Dr. Jeff Toth

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Abstract

The current study uses the SportGait BioKinetoGraph (BKG), a validated mobile application based on accelerometer technology, to investigate the effect of physical fatigue on gait dynamics such as speed, power, balance, and symmetry. Building on research linking fatigue to increased injury risk, particularly concussions, this study investigates how physical fatigue influences gait. Using a within-group experimental design, gait metrics were measured before and after exercise in participants recruited from a university recreation center. Subjective and objective assessments included self-reported (SR) fatigue and heart rate (HR) assessed both prior to and after the participant's usual workout. The study hypothesizes that fatigue-induced gait changes will result in longer stride time (slower speed), less power, and decreased balance. Using paired samples t-tests the findings indicate that, although fatigue has an impact on gait performance, the effects are small and domain-specific, with changes noted primarily on stride and power parameters. Pearson correlations also show some modest associations between the BKG variables and SR fatigue, and to a lesser extent HR. The SR and HR measures were not significantly correlated, suggesting these outcomes capture distinct aspects of fatigue. The absence of broader findings may have been affected by the small sample size, which limits power. These findings highlight the potential utility of differentiate between fatigue-related gait alterations and those caused by mild traumatic brain injuries (mTBI) in clinical and athletic contexts.

Keywords: physical fatigue, gait metrics, self-reported fatigue, heart rate

The nature of fatigue

Fatigue is a complex, multifaceted symptom linked to numerous medical conditions commonly described as exhaustion or reduced energy that impairs voluntary activity (Gruet et al., 2013). It manifests as either mental fatigue, involving cognitive or perceptual exhaustion, or physical fatigue, reflecting reduced motor system performance (Kunasegaran et al., 2023) or some combination of the two.

Physical fatigue occurs when muscles fail to maintain strength during exertion due to central or peripheral mechanisms affecting the nervous system and musculature (Abd-Elfattah et al., 2015). Central fatigue originates in the central nervous system (CNS), where weakened neural signals to muscles reduce motor drive (Abd-Elfattah et al., 2015). This process is tied to biochemical changes in neurotransmitters such as serotonin, dopamine, and norepinephrine, which regulate motivation, arousal and motor activity. Elevated serotonin levels, relative to dopamine, are particularly associated with fatigue and diminished motor motivation (Tornero-Aguilera et al., 2022). Whereas peripheral fatigue develops at or beyond the neuromuscular junction, often due to depleted energy substrates (e.g., adenosine triphosphate, glycogen) or the accumulation of metabolic byproducts that impair calcium handling and muscular contraction (Wan et al., 2017; Keyser, 2010). These disruptions reduce the muscles' capacity to generate force, limiting performance.

Fatigue can also extend beyond physical symptoms, encompassing mental and emotional exhaustion, often resulting from sustained cognitive effort, illness, or prolonged activity. Mental fatigue is characterized by reduced cognitive function, diminished motivation, and a heightened perception of effort (Meeusen et al., 2020). Biochemically, it mirrors central fatigue, with

changes in brain metabolism and elevated adenosine levels, which promote sleep and reduced arousal (Meeusen et al., 2020). A notable study found that mental fatigue limits exercise tolerance by increasing perceived effort rather than directly impairing muscular or cardiovascular systems (Marcora et al., 2009). In this respect, mental and physical fatigue are interconnected processes.

Despite its prevalence, fatigue remains challenging to define due to its diverse manifestations. Researchers have developed frameworks to address its various components, such as reduced muscular efficiency, slower responsiveness, and decreased motivation. A unifying characteristic, however, is its reversibility and its short-term impact on skeletal muscle performance, often reflected in decreased force or power output (Marino, 2020). Understanding these mechanisms is critical for exploring the broader implications of fatigue on human performance and its role in clinical and athletic contexts.

Fatigue in concussion management

Understanding fatigue is critical to effectively diagnose and treat illnesses and injuries such as concussions. Baseline ratings on concussion-related symptom assessments are critical for guiding clinicians' return-to-play choices. Symptoms typically reported in baseline examinations, such as fatigue, headache, and difficulty concentrating, are equally widespread post-concussion, underlining the importance of establishing correct baseline rates for these symptoms (Mrazik et al., 2013). Moreover, assuming most concussions occur during the course of play, fatigue is often an unquantified aspect of any assessment that can impact performance. Concussions require a thorough examination that includes motor control, neurocognitive function, and self-reported symptoms (Scorza et al., 2012). Persistent deficiencies in these regions can have long-

term consequences, such as depression and cognitive deterioration. Repeat concussions raise the likelihood of future injuries, necessitating cautious return-to-play measures and annual baseline monitoring (McAllister & McCrea, 2017).

Baseline assessments serve as personalized benchmarks for comparing post-injury conditions; however, fatigue levels can impact their accuracy during testing (Combs et al., 2019), especially when fatigue levels are not comparable post-injury relative to baseline testing. Fatigue complicates the diagnosis of concussion because it can impact neurocognitive performance, postural control, and symptom reporting. Thus, including fatigue-related information and characteristics in concussion management is critical to ensure proper evaluations and interventions (Nardone et al., 2021).

Fatigue and physical performance

Fatigue greatly impacts performance in sports and makes individuals more susceptible to injury. For example, approximately 3.8 million concussions occur each year as a result of recreation and sports activities, with collegiate athletes accounting for a sizable fraction of the total (Combs et al., 2019). According to studies, players' risk of concussion increases as they spend more time on the playing surface (e.g., ice-time during hockey games), emphasizing the importance of fatigue in injury risk (Stevens et al., 2008). The reason for this is that fatigue can slow reaction times, making it difficult for athletes to avoid accidents and increasing the risk of concussions (Wylie & Flashman, 2017). Kinchington et al., 2010 found evidence that higher levels of perceptual fatigue are associated with an increased risk of injury. Specifically, they observed that lower scores on the Lower-Limb Comfort Index (LLCI) [124], indicating greater perceptual fatigue, were correlated with a higher incidence of lower-limb injuries ($r = 0.88$; $p < 0.001$) among elite contact-sport athletes. For instance, rugby players who underwent high-

intensity fatigue protocols experienced decreases in power, velocity, and jump height after exercising. Fatigue has a negative impact on balance and stability, as evidenced by vertical jump tests that revealed a reduction in leap height and balance among fatigued individuals (Aquino et al., 2022). This discovery is consistent with the stress and injury model forwarded by Williams and Andersen, which indicates that elevated stress levels and insufficient coping mechanisms increase the likelihood of sustaining an injury (Williams & Andersen, 1998).

Understanding balance and fatigue

Maintaining balance and an upright posture is a complex process that involves many different body systems. Although muscles play an important function, they are insufficient to prevent loss of balance when external forces occur on the body. Instead, our bodies rely significantly on the central nervous system (CNS) to coordinate and regulate movements, ensuring the appropriate muscles are engaged to maintain stability (Stodolka et al., 2016). Fatigue is another important aspect in determining balance. It is an intrinsic disorder that impairs the central nervous system's ability to effectively regulate muscular action. The CNS consists of the brain and spinal cord, with higher motor neurons from the motor cortex transmitting impulses to the spinal cord. These impulses activate the lower motor neurons at the neuromuscular junction (NMJ), resulting in muscle contraction (Tornero-Aguilera et al., 2022).

This concept of the fatigue response can also be understood as a controlled process that operates within a series of interconnected biological structures, akin to a "chain." The lower motor neurons at the NMJ, sometimes referred to as motor nerve fibers, are what are commonly meant when the phrase "motor nerve" is used (Marino, 2020). In a study done by Nardone et al. (2021) the effects of fatigue on balance while standing still were tested. The results showed that multiple factors affected how physical exercise affects the body sway of young, healthy

participants. Muscle fatigue causes significant alterations in gait, including greater speed and longer stride length, which indicates a more energetic but less steady walking pattern. These changes may arise from a compensatory mechanism in which less affected muscles engage more intensely to sustain movement (Santos et al., 2019). The impact of fatigue on walking patterns can also differ based on the conditions. Walking on the ground tends to result in more noticeable fatigue effects in comparison to walking on a treadmill. The difference is likely because walking on a natural surface (e.g., outside on grass) involves a greater range of natural and unpredictable circumstances, necessitating constant modifications and adjustments. In contrast, walking on a treadmill takes place in a consistent and controlled setting, reducing the necessity for such adjustments (Santos et al., 2019). Objectively assessed gait parameters (including balance) have been shown to differ when comparing walks on a man-made, hard surface to walks on a natural surface such as grass (Lecci, Williams, et al., 2023). Research conducted by Santos and colleagues also demonstrates that the kind of exercise (cycling or walking on a treadmill), the level of exercise intensity (above or below the anaerobic threshold), and the visual conditions had an impact on the functional imbalance of individuals (Santos et al., 2019). Walking on a treadmill increased body sway (destabilization) when the exercise was exhausting (above the anaerobic threshold), but not when it was below the anaerobic threshold (Nardone et al.,2021). A study that was investigating the impact of fatigue in the muscles of the lower limbs in patients diagnosed with Parkinson's Disease (PD) found that lower limb fatigue has a considerable impact on spatiotemporal gait characteristics and turning performance (Halimatul Abd Ghani et al., 2023). The degree of instability was controlled by the visual environment, with eyes open producing lower increases in body sway. Findings also suggest that exhausting workouts may cause metabolic and excitation-contraction coupling problems, which might increase body sway.

Testing showed that the frequency of breathing was, on average, only 20% higher in the first block of trials following the exhausting treadmill and cycle ergometer exercises. As a result, one would anticipate only very slight variations in body sway following the exercises solely due to respiration. It is important to note that according to Nardone and colleagues (2021), all of the immediate and transient effects of exercise on body sway vanished 15 minutes after exercise. Another study showed substantial alterations in various spatio-temporal gait metrics after the training (Shin et al., 2024). The cadence, which refers to the frequency of steps done, exhibited a substantial decrease.

The study found a notable reduction in the proportion of the single support phase and a noteworthy increase in the proportion of the double support phase stance. These findings indicate that after exercising, participants tended to rely more on both feet for support, most likely because their stability had diminished. In addition, there was a notable rise in toe-off time and gait asymmetry, suggesting a decline in gait stability and coordination after exercising. However, there was no significant difference in the normalized stride length (Shin et al., 2024).

Fatigue and Mild Traumatic Brain Injury (MTBI)

Mild Traumatic Brain Injury (MTBI), often known as a concussion when discussed within the context of sports, affects many individuals each year, with occurrence rates ranging from 100 to 300 per 100,000 people (Cassidy et al., 2004). Although most patients recover within three months after the injury, a significant subgroup continues to experience persistent symptoms beyond this timeframe (McInnes et al., 2017). Out of all the concussion symptoms, fatigue is especially common and can last for a long time after the initial injury, thereby making the recovery process and daily activities more difficult (Stulemeijer et al., 2006). Fatigue is a prevalent symptom that affects patients' concentration, drive, and physical activity levels, both in

the short term and in the long term. This makes it a crucial aspect to consider when managing MTBI (Borgaro et al., 2005).

Baseline examinations are crucial in the management of concussions since they establish an individualized standard for comparing the conditions after an injury. Obtaining precise first data on symptoms such as fatigue, headache, and difficulty concentrating is essential for making informed judgments about whether it is safe to resume playing sports (Mrazik et al., 2013). Baseline data can aid in differentiating between pre-existing issues and those that directly stem from head trauma. Nevertheless, the variable nature of fatigue might affect the dependability of initial measurements, making the evaluation process more complex when fatigue levels after an injury are significantly different from the baseline conditions (Combs et al., 2019).

In addition to making individuals more prone to injury, the detection and treatment of concussions are made considerably more challenging by fatigue. Fatigue can negatively impact cognitive function, disrupt the ability to maintain proper posture, and alter the reporting of symptoms, thereby complicating the appropriate assessment of the patient's state (Nardone et al., 2021). Fatigue may hide other symptoms of a concussion (e.g., if concussion symptoms are inaccurately attributed to fatigue) or exaggerate their severity (if fatigue exacerbates the presentation of other concussion symptoms), with either scenario potentially resulting in misdiagnosis or insufficient treatment. Incorporating fatigue evaluations into conventional concussion management protocols is crucial for improving the thoroughness of assessments and increasing the potential for successful interventions (Silverberg et al., 2015). Individuals with mild traumatic brain injury (MTBI) may have persistent post-concussive symptoms that impact their physical and social capabilities. Stulemeijer et al. (2006) discovered that fatigue, in addition to problems with focus and decreased motivation, has a considerable negative impact on the

overall well-being and everyday functioning of patients with MTBI. Their research uncovered that approximately 33% of patients with MTBI suffered profound fatigue six months after the accident, in contrast to only 12% of patients with mild orthopedic injuries. The results emphasize the significant impact of fatigue in individuals with MTBI (Stulemeijer et al., 2006).

Of course, fatigue is a common experience among the general population. According to epidemiological research, between 9.75% and 33% of persons experience brief periods of fatigue lasting less than a month, with the majority of estimates falling between 15% and 25% (Jason et al., 2010). Non-pathological fatigue is the term used to describe the typical and anticipated feeling of fatigue that follows physical exercise or prolonged activity. In athletes, this form of fatigue is a prevalent and inherent reaction to training and competition (Moeller, 2004). It is usually short-lived and can also be caused by such things as a lack of sleep or transitory sickness, whereas pathological fatigue is related to persistent health issues and can severely impede everyday functioning and well-being (Jason et al., 2010). In the current study, we will focus on non-pathological fatigue, as it is easier to quantify and assess experimentally.

Introduction of a BioKinetoGraph (BKG) for the assessment of gait

As noted in the literature review, fatigue can have a significant impact on balance and stability, leading to increased postural sway and decreased control over movements, as well as decreased power and speed. Assessing these variables using the BKG system, which has been validated for use in post-concussion evaluations, could provide insights into the effects of fatigue on gait stability.

The BKG can be deployed through an app on any mobile device, and the mobile assessment automatically produces T-scores based on normative data for a number of gait variables. Similar to how an electrocardiogram depicts heart activity, BKG waveforms visually

represent specific neuromotor gait cycle events such as cadence, foot strike, push-off, double stance time, and swing phase. The BKG system utilizes raw data from accelerometers to generate waveforms representing gravitational accelerations over time. These waveforms are then analyzed, and algorithms are used to extract information related to the range of motion, amplitudes, and timing intervals of various gait components. The data collection method aligns with similar studies published in the field (e.g., Lecci, Dugan, *et al.*, 2023). Although gait data can be collected from sensors attached to different parts of the body, most published research focused on data derived from a single sensor affixed to the sacrum or held against the sternum. This approach records inertial measurement units (IMUs) via transmitted signals to a local device and then to remote servers for BKG motion analysis. It is at that point that algorithms process the raw data to detect critical gait cycle markers and extract spatial, temporal, kinetic, and spectral features (Lecci, Williams, *et al.*, 2023).

This study focuses on a subset of BKG variables that have been shown to be reliable and valid measures of gait (Keith et al., 2019; Lecci, Dugan, et al., 2023; Lecci, Williams et al., 2023) and are organized into four conceptual domains: balance (stability), stride (timing intervals), power (amplitudes), and symmetry (regularity and consistency). These domains encompass various aspects of gait and are represented by 18 specific variables (see Table 1, which provides the variable names and their respective domains).

Hypothesis

This study proposes that although fatigue can disrupt neuromotor performance and introduce variability into assessment results, it does not independently reproduce concussion symptoms. Instead, a fatigue-specific gait signature can be distinguished from a concussion-defined gait signature. To investigate this, a within-group experimental design is utilized to

assess dynamic gait before and after a fatigue-inducing workout. The BioKinetoHraph (BKG) system is used to analyze key gait characteristics offering insights into the impact of fatigue on gait. Specifically, it is hypothesized that following the participants' workout, their gait scores will reflect reduced power (force), slower speed (longer stride times), decreased balance (stability), and less symmetry. Paired-samples t-tests will be used to test this hypothesis. It is also anticipated that these effects will be stronger as a function of greater fatigue, as assessed by self-reported fatigue and a heart rate (HR) measure. This hypothesis will be tested using Pearson correlations between the BKG variables and the HR and self-report measures of fatigue. Finally, it is expected that these effects will lessen over time after the workout is completed, capturing the recovery process. This can be tested by repeating the Pearson correlations with time since workout as a covariate (i.e., a partial correlation).

Validating fatigue-specific gait patterns has critical implications for clinical and athletic settings. Current concussion assessments often overlook objectively considering the confounding effects of physical fatigue, which can mimic concussion symptoms, leading to misdiagnosis or premature return-to-play decisions. For instance, fatigue can reduce postural stability and motor control, both of which are also key indicators of concussion-related impairments. By establishing a clear distinction between these patterns, clinicians can refine their diagnostics criteria, improving the accuracy of baseline and post-injury evaluations. Ultimately, this research can contribute to a deeper understanding of how fatigue influences neuromotor performance and support the development of tailored concussion assessments that account for those effects

Method

Participants

Individuals were recruited from the UNCW Recreation Center. To qualify for the study, individuals had to be 18 years of age or older and had to be able to ambulate independently. The sample included 58.18% men, 40% women, and 1.82% non-binary people. The participants' ages ranged from 18 to 64 years, with a mean age of 20.51 years (*S*=4.98, SD=4.98). In terms of racial makeup, the majority of participants (86.24%) identified as White, followed by African American (7.34%), Other (3.67%), Asian or Pacific Islander (0.92%), and Native American (0.92%). Sixty-nine participants completed all of the pre-workout measures, and 52 participants completed the post-workout measures, indicating 24.6% participant loss.

Measures

The researchers utilized the Multidimensional Fatigue Inventory (MFI) created by Smets et al. (1995) to evaluate subjective fatigue levels in the areas of physical, cognitive, and psychosocial aspects. Responses were on a 5-point Likert scale, and higher scores indicated more fatigue. This measure had been tested in a variety of populations, including healthy people and those with chronic illnesses, displaying high reliability and construct validity. Its multidimensional structure enabled a thorough assessment of fatigue, distinguishing between various forms such as physical fatigue, mental fatigue, and decreased motivation. By using this method, the researchers aimed to capture nuanced variations in fatigue that resulted from the experimental conditions.

A laboratory phone equipped with the BKG application measured the gait parameters organized into four domains after the participants completed their workouts. A Garmin watch assessed the heart rates of participants both prior to and following workout sessions.

BKG

There are 4 BKG domains: power, symmetry, balance, and stride. The balance domain evaluates walking stability by analyzing the range of motion (ROM) at the center of mass (sacrum) and the variability within the gait cycle. In this context, stability refers to a person's level of steadiness when carrying out various walking-related tasks (e.g. Side balance is a metric that evaluates stability when walking in a straight line during the gait cycle). Turnaround balance refers to the ability to maintain stability during the turnaround phase of the gait cycle. It is an important measure for assessing an individual's ability to maintain balance when changing direction. Sway time is a metric that quantifies the duration of side-to-side motions, offering valuable information on one's capacity to maintain balance in the lateral direction. Side power refers to the force produced during lateral motions, which reflects the level of strength and control in side-to-side motion. Stride smoothness indicates the regularity of walking patterns, indicating the level of smoothness in an individual's stride. Consistency support quantifies the degree of variation in the duration of the double stance phase, during which both feet are in touch with the ground. The values in the Balance domain are inverted, indicating that lower scores imply difficulty functioning. This inversion simplifies the understanding of data, enabling the identification of places where stability may be jeopardized (Lecci et al., 2023). The stride domain measures the duration required to complete various stages of the gait cycle, effectively acting as a substitute for speed. The measurement encompasses multiple crucial factors that assess various facets of stride consistency and timing. Stride time is the average time interval

between when the heels of opposite feet hit the ground, which gives a general indication of how consistent a person's stride is. Double stance refers to the duration in which both feet are in touch with the ground. It provides information on the average time spent in this phase. The stance phase quantifies the mean duration of right and left side stances, reflecting the degree of weight distribution during walking. The swing phase measures the mean duration of the movement when an individual propels each foot forward, indicating their efficiency in doing so. Similar to balance values, all stride values are reversed, where lower scores indicate worse functioning (i.e., a slower walking pace). Standardization simplifies the process of comparing individuals and judgments (Lecci et al., 2023). The symmetry domain quantifies the uniformity and consistency of motion in the anterior-posterior and vertical dimensions of walking. This domain encompasses crucial characteristics that offer insights on the uniformity of movements during walking. Forward movement symmetry is a measure that assesses the balance between forward and backward displacement from the center of mass. It indicates the degree to which gait movements are uniformly distributed in both the forward and backward directions. Vertical movement symmetry quantifies the extent of upward and downward displacement relative to the center of mass, offering valuable information on the consistency of vertical gait. Vertical sway symmetry is a measure of how evenly the upward and downward motions from the center of mass occur, indicating the consistency of vertical movements. Low scores in the symmetry domain indicate impaired functioning, emphasizing differences that may indicate neuromotor problems (Lecci et al., 2023). The power domain represents the magnitude of walking, which is determined by the quantity and effectiveness of energy produced and used throughout the walking cycle. This domain has multiple essential variables that assess various aspects of the energy dynamics associated with walking. Total power quantifies the total resultant force exerted at the center of

mass, representing the comprehensive use of energy during the act of walking. Vertical power measures the energy produced during vertical movements, reflecting the effectiveness and potency of upward and downward actions. Forward power quantifies the exertion generated during forward motions, indicating the amount of energy utilized to move the body in a forward direction. Striking force quantifies the force produced by the foot strike, reflecting the magnitude of impact with each step. The pushing force measures the force produced when the toe pushes off, indicating the effectiveness of the last stage of the walking cycle. Low scores in the Power domain indicate impaired functioning, indicating problems with the production and use of energy during walking (Lecci et al., 2023).

Procedure

Upon reaching the Student Recreation Center, potential subjects were approached and asked about their willingness to participate in a study focused on understanding fatigue. Individuals who were interested provided informed consent and filled out a demographic survey, as well as the MFI, to establish their baseline fatigue levels. A laboratory phone equipped with the BKG application measured the gait parameters organized into four domains. The values to be examined were the mean of the values from the 2nd and 3rd walks, both pre- and post-workout. The resting heart rate of each participant was assessed using a Garmin watch. Participants subsequently engaged in three walks, during which they held a phone against their chest with the screen facing inward. The walks were conducted without shoes or with socks to ensure uniformity of circumstances and consistency among all participants. Participants performed their usual exercise regimens at the Student Recreation Center, selecting their chosen activities, which involved aerobic workouts, resistance training, or flexibility exercises. Individuals determined the length and level of difficulty of their exercise session according to their specific tastes and

regular workout patterns. Following the completion of their workout, the participants' heart rates were assessed again utilizing the Garmin watch. After their workout, participants filled out a post-workout survey. This survey asked questions regarding the length and type of their activity, the amount of time that had passed since they finished exercising, and the MFI (Multidimensional Fatigue Inventory) to evaluate their fatigue levels after the workout. Finally, participants engaged in three additional walks that were identical to the initial walk test. During these walks, they either walked barefoot or in socks while holding the phone against their chest with the screen facing inward.

Results

A paired-samples t-test was used to assess changes in self-reported fatigue scores between pre ($M = 52.7$, $SD = 7.42$) and post-workout ($M = 97.6$, $SD = 10.5$). The results showed a significant difference in pre- and post-fatigue scores ($t(48) = -51.4$, $p < .001$), with a mean difference of 44.9 ($SE = 0.873$), indicating a large effect size (Cohen's $d = 7.34$). This shows that participants reported significantly more fatigue following the workout as compared to pre workout scores.

Heart rate, on the other hand, did not differ significantly between pre- ($M = 78.6$, $SD = 10.6$) and post-workout ($M = 80.5$, $SD = 14.5$) measures, $t(56) = -0.583$, $p = .562$, with a mean difference of -1.47 ($SE = 2.53$). These findings indicate that the self-paced exercise did not result in a significant change in heart rate, or that heart rate was insensitive to the effects of the workout. The absence of a significant change from pre to post HR values may be due to individual variability in fitness and recovery abilities or inadequate exertion to produce a quantifiable physiological response.

The self-reported fatigue and HR measures were also correlated with each other to determine if there is any convergence between the two outcomes. Pearson correlation analyses were performed independently on pre- and post-workout variables. For pre-workout measurements, there was no significant relationship between self-reported fatigue scores and HR ($r = 0.047$, $df = 46$, $p = 0.751$). For post-workout measures, the correlation between HR and self-reported fatigue scores was also not significant ($r = 0.046$, $df = 41$, $p = 0.770$). This shows there is no significant association between perceived fatigue and heart rate prior to or after the workout.

These data indicate that self-reported fatigue and HR may reflect different components of physiological and subjective fatigue sensations, with no evidence of overlap between the two measures. Moreover, follow-up analyses on other variables collected, including self-reported strenuousness of the workout and time since completion of the workout, were also unrelated to self-reported fatigue or HR.

The lack of substantial relationships between all of these variables indicates that subjective feelings of workout difficulty and self-reported fatigue are not highly related to objective performance metrics or physiological fatigue. These findings underscore the complexities of fatigue, as psychological and physiological components may function independently, representing distinct aspects of the exertion experience.

The effect of participant workout

To test the first hypothesis, paired-samples t-tests were used to compare pre- and post-workout walks using various gait parameters assessed by the SportGait BioKinetoGraph (BKG). The results are presented in Table 1.

Significant increases were observed in Swing Phase ($t(51) = 2.045$, $p = .046$, $d = 0.284$) and Stride Time ($t(51) = 2.06$, $p = .04$, $d = 0.29$). The mean differences (6.74 ms and 7.75 ms, respectively) indicate that participants spent more time in the swing phase (i.e., walked slower) following the activity. These data show that fatigue produces noticeable temporal changes in gait (slowed gait), which is consistent with predictions but has small effects.

Vertical power decreased considerably between the first and second walks ($t(51) = -2.347$, $p = .023$, $d = -0.326$), with an average difference of -0.048. This suggests that participants' vertical force generation decreased after the activity.

Forward Power ($t(51) = -0.422$, $p = .675$) and Total Power ($t(51) = 0.331$, $p = .742$) remained consistent between conditions. There was a trend toward a decrease in Side Power ($t(51) = -1.694$, $p = .096$) following exercise, with a mean difference of -0.017. Although not statistically significant, this change is consistent with the predicted pattern, indicating diminished power after exercise; however, the impacts appear to be small.

The stability metrics revealed no significant changes in Double Stance ($t(51) = 0.095$, $p = .925$) or Support Variability ($t(51) = -1.881$, $p = .066$), with the latter approaching significance. Side Sway also approached significance ($t(51) = -1.954$, $p = .056$), with a mean difference of -0.003. This indicates that sway variability and sideway may increase following strenuous activities, which is consistent with predictions, but the effects are small and do not meet statistical significance.

The symmetry measures revealed minor trends but no substantial changes. During the second walk, both Forward Movement Symmetry ($t(51) = -1.967$, $p = .055$) and Vertical Movement Symmetry ($t(51) = -1.609$, $p = .114$) showed potential symmetry loss. Vertical Sway

Symmetry remained stable over walks ($t(51) = -0.274$, $p = .785$). Gait Smoothness ($t(51) = 1.269$, $p = .210$) was similar throughout walks.

When only participants with a higher HR post-workout were studied, significant changes were detected in temporal gait parameters (e.g., swing phase, stride time), but not in power, stability, or symmetry. These findings imply that people with high HR post workout, experience fatigue-related changes in the timing of their gait rather than its stability or power components, though it should be noted that the smaller sample size will inevitably reduce the number of significant findings due to decreased power.

The association between BKG gait metrics and fatigue indicators

To address hypothesis two, the relationships between gait metrics, heart rate (HR), and self-reported fatigue (SR Fatigue) were analyzed using Pearson correlations in order to determine if the extent of the fatigue relates to gait. In the power domain, Side Power has a significant negative correlation with SR Fatigue. Pre-fatigue Side Power correlates with post-fatigue SR Fatigue ($r = -0.285$, $p < .05$), while post-fatigue Side Power correlates with both pre-fatigue ($r = -0.294$, $p < .05$) and post-fatigue SR Fatigue ($r = -0.284$, $p < .05$). Pre-fatigue Vertical Power was significantly negatively correlated with HR Pre ($r = -0.390$, $p < .01$), whereas post-fatigue Vertical Power was also negatively correlated ($r = -0.294$, $p < .05$).

In the stride domain, Stride Time had a significant negative correlation with pre-fatigue SR Fatigue Pre ($r = -0.297$, $p < .05$), suggesting it may be sensitive to perceived fatigue. Post-fatigue Stride Time followed a similar pattern, however the correlation was not statistically significant ($r = -0.246$), likely due to the smaller sample size after the workout (i.e., not all participants returned for the second assessment). There were no significant correlations between

Double Stance, Stance Phase, and Swing Phase and HR or SR Fatigue under pre- and post-fatigue conditions.

The stability domain uncovered more subtle interactions. Side Sway pre-fatigue had a strong negative correlation with SR Fatigue Pre ($r = -0.360$, $p < .05$). Significant negative associations were found between pre-fatigue Turn Around Sway and both SR Fatigue Pre ($r = -0.289$, $p < .05$) and SR fatigue Post ($r = -0.294$, $p < .05$), indicating that more sway (i.e., imbalance) is related with higher fatigue. Post-fatigue Turn Around Sway showed a positive correlation with HR Post ($r = 0.323$, $p < .05$), suggesting a physiological response to fatigue. Similarly, Sway Velocity Post had a strong positive connection with HR Post ($r = 0.366$, $p < .05$), indicating its sensitivity to physical activity. However, there was no significant correlation between Support Variability and HR or SR Fatigue.

In the symmetry domain, Vertical Movement Symmetry Pre workout was negatively associated with HR Pre workout ($r = -0.279$, $p < .05$), indicating a link between vertical symmetry and physiological exertion. Vertical Sway Symmetry had a strong positive connection with HR Pre and SR pre-fatigue ($r = 0.331$, $p < .01$) and post-fatigue ($r = 0.359$, $p < .01$) circumstances, indicating greater movement (imbalance) occurs when individuals are fatigued. However, Forward Movement Symmetry had no significant relationships with HR or SR Fatigue.

Determining the association between BKG gait metrics and fatigue indicators after controlling for time since ending the workout

To address the third hypothesis, partial correlations were used to see whether the time since workout completion affected the connection between fatigue ratings and BKG scores. Pearson partial correlations were employed, with time since workout as the covariate. The

analysis found no significant difference in the partial correlations compared to the original Pearson correlations. These findings indicate that the relationship between fatigue ratings and BKG scores is unaffected by time since completion of the workout, illustrating the robustness of the initial findings.

Exploratory analyses

To examine the relationship between subjective perceptions of workout difficulty and objective performance on the BKG, Pearson correlations were conducted. Overall, there were no statistically significant association between the BKG measures and self-reported strenuousness of the workout. These findings indicate that participants' subjective judgments of workout difficulty do not correspond to their gait as objectively assessed with the BKG.

Table 1. Paired Samples t-test BKG variables

Domain	Variable	Pre M(SD)	Post M(SD)	t-value	df	p	Mean Difference	SE Difference	Effect Size (Cohen's d)
Power	**Forward Power**	0.43 (0.08)	0.41 (0.12)	-0.42	51	0.68	-0.01	0.01	-0.06
	Pushing Force	2.42 (0.92)	2.34 (0.9)	-1.37	51	0.18	-0.12	0.08	-0.19
	Side Power	0.354 (0.11)	0.364 (0.1)	-1.69	51	0.10	-0.02	0.01	-0.23
	Striking Force	3.36 (0.95)	3.25 (0.9)	-1.15	51	0.26	-0.12	0.10	-0.16
	Total Power	0.380 (0.12)	0.37 (0.14)	0.33	51	0.74	0.004	0.01	0.05
	Vertical Power	0.836 (0.29	0.82 (0.32)	-2.35	51	0.02*	-0.05	0.02	-0.33
Stride	**StancePhase**	683 (61.6)	681 (61.8)	1.37	51	0.18	6.08	4.45	0.19
	StrideTime	583 (54.9)	579 (50.7)	2.06	51	0.04*	7.75	3.76	0.29
	SwingPhase	491 (46.7)	487 (44.0)	2.05	51	0.05*	6.74	3.29	0.28
	DoubleStance	98.2 (10.8)	99.3 (11.2)	0.1	51	0.93	0.12	1.21	0.01
Stability	**Gait Smoothness**	2.29 (0.75)	2.25 (0.94)	1.27	51	0.21	0.12	0.09	0.18
	Side Sway	0.04 (0.02)	0.04 (0.02)	-1.95	51	0.06	-0.003	0.00	-0.27
	Support Variability	12.1 (9.26)	13.1 (5.74)	-1.89	51	0.07	-2.68	1.43	-0.26
	Sway A	0.03 (0.01)	0.02 (0.01)	0.058	51	0.95	4.72	0.00	0.01
	Sway Velocity	0.32 (0.11)	0.33 (0.1)	-0.55	51	0.58	-0.003	0.01	-0.08
	Turn around sway	0.06 (0.03)	0.07 (0.02)	-0.07	51	0.94	-0.0002	0.00	-0.01
Symmetry	**Forward Movement Symmetry**	-40.0 (19.0)	-35.0 (19.1)	-1.97	51	0.06	-4.82	2.45	-0.27
	Vertical Movement Symmetry	-11.7 (18.6)	-11.9 (17.5)	-1.61	51	0.11	-2.1	1.30	-0.22
	Vertical Sway Symmetry	21.6 (9.61)	22.0 (11.6)	-0.27	51	0.79	-0.23	0.86	-0.04

***Note**:* Table displays the results of paired samples t-tests comparing pre- and post-measures for various BKG variables across domains (e.g., power, stride, stability, symmetry). Reported values include pre- and post-means with standard deviations (M ± SD), t-values, degrees of freedom (df), p-values, mean differences, standard errors of the differences (SE Difference), effect sizes (Cohen's d), and power estimates.

Table 2. Pearson Correlations between BKG and Outcome Variables

Domain	BKG Variables	HR Pre	HR Post	SR Fatigue Pre	SR Fatigue Post
N		42	35	27	27
Power	Forward Power Pre	-.082	.064	.101	.053
	Forward Power Post	-.117	.109	.067	.011
	Total Power Pre	-.093	.175	0.047	-0.046
	Total Power Post	-.188	.044	.101	.07
	Side Power Pre	-.062	.136	-0.270	-0.285*
	Side Power Post	-.026	.230	-.294*	-.284
	Striking Force Pre-Workout	-.237	-.004	.123	.08
	Striking Force Post-Workout	-.130	.095	.024	-.015
	Vertical Power Pre	-0.390**	-.083	0.211	0.128
	Vertical Power Post	-.294*	-.013	.237	.118
	Pushing Force Pre-Workout	-.194	.082	-.083	-.109
	Pushing Force Post	-.162	.182	.022	-.015
Stride	Double Stance Pre	-0.090	-.002	-0.069	-0.001
	Double Stance Post	.039	.022	–.127	0.0
	Stance Phase Pre	.136	.104	-.249	-.152
	Stance Phase Post	.135	.113	-.214	-.129
	Swing Phase Pre	.146	.167	-.277	-.188
	Swing Phase Post	.141	.165	-.234	-.176
	Stride Time Pre	.139	.134	-.297*	-.207
	Stride Time Post	.119	.173	-.246	-.158
Stability	Support Variability Pre	-.077	.023	-.230	-.165
	Support Variability Post	.061	.009	.118	-.154
	Side Sway Pre	.006	.131	-.360*	-.247
	Side Sway Post	.126	.276	-.254	-.196
	Turn Around Sway Pre	-.029	.110	-289*	-.294*
	Turn Around Sway post	.074	.323*	-.263	-.247
	Sway Velocity Pre	.002	.182	-.197	-.211
	Sway Velocity Post	-.006	.366*	-.248	-.215
	Gait Smoothness Pre-Workout	.227	-.084	.081	.121
	Gait Smoothness Post-Workout	.091	.024	.142	.187
	Sway A Pre-Workout	-.150	.286*	-.141	-.133
	Sway A Post-Workout	.014	.328*	-.083	-.133
Symmetry	Forward Movement Symmetry Pre	-.068	.077	.095	0.04
	Forward Movement Symmetry Post	.105	-.054	.066	-.057
	Vertical Movement Symmetry Pre	-.279*	-.068	.222	.18
	Vertical Movement Symmetry Post	-.151	.011	.143	.062
	Vertical Sway Symmetry Pre	.331**	.012	-.093	.034
	Vertical Sway Symmetry Post	.359**	.029	-.058	-.033

Note: * $p < .05$, ** $p < .01$, *** $p < .001$.

Discussion

This study investigates the effects of fatigue on gait metrics, confirming the hypothesized direction with significant but small findings. Notably, SR fatigue increased significantly post-exercise, heart rate (HR) remained stable, and most gait metrics exhibited only minor changes. The BKG effects of fatigue, although significant, may have been influenced by the variability in the participants' workout intensities and recovery time. The modest sample size, especially post workout, likely contributed to several of the findings failing to achieve statistical significance due to lower power.

The weak correlation between SR fatigue and HR suggests that these measures assess different dimensions of fatigue. HR may reflect acute physiological exertion, whereas SR fatigue likely encompasses a combination of long-term physical and psychological factors. Interestingly, HR data showed no substantial increase post-workout, which contradicts conventional exertion-related reactions. Participants may have used caffeine-rich pre-workout supplements, which could explain some of these effects, thereby increasing their resting heart rate prior to the initial assessment. In such circumstances, the subsequent post-workout HR may have appeared lower because participants returned to a state closer to their baseline when activity ended, rather than representing a lack of exertion. This explanation emphasizes the significance of accounting for pre-exercise activities when analyzing HR data.

Changes observed in gait from pre to post workout that related to fatigue predominantly occurred in the stride and power domains, with notable increases in swing phase and stride time. These findings imply that participants may have adopted slower, more deliberate steps following their workouts, potentially as a compensatory response to fatigue. This aligns with previous

research stating that fatigue can cause changes in gait patterns, such as increased stride frequency and decreased stride length, as a means of maintaining stability and conserving energy. For example, Hak et al. (2013) discovered that raising stride frequency improves medio-lateral margins of stability, implying that people may take shorter, faster steps to improve balance while fatigued. Similarly, Fukuchi et al. (2019) found that walking speed influences spatiotemporal gait parameters, with greater speeds resulting in shorter stride lengths and higher frequencies, possibly as a compensatory response to fatigue.

The significant reduction in vertical power observed in this study suggests decreased force generation, consistent with neuromotor inefficiency induced by fatigue. This is consistent with the findings of Morales-Artacho et al. (2017), who discovered that fatigue procedures caused selective decreases in the maximal capacities of upper body muscles to produce force, velocity, and power. The study found that varied fatigue procedures resulted in significant reductions in power production during multi-joint tasks, emphasizing the importance of fatigue in neuromuscular function. Similarly, Enoka and Duchateau (2008) discussed how muscle fatigue might impair muscular function and joint stability, reducing exercise performance. They observed that muscle fatigue is connected with a brief decrease in maximum muscle force or power caused by exercise, which can be attributable to a variety of processes, including metabolite accumulation inside the muscle and insufficient motor command in the neuromuscular system. While changes in stability and symmetry metrics were minor, some measures (such as side sway and forward movement symmetry) neared significance, indicating subtle tendencies that require further research.

These findings demonstrate that fatigue impairs the neuromuscular system's ability to generate force efficiently, resulting in reduced power production and movement effectiveness.

Furthermore, they propose that when fatigued, people adjust their stride by taking shorter, faster steps in order to conserve energy and maintain stability.

Interestingly, despite a significant increase in self-reported fatigue following exercise, several participants reported feeling more fit or revitalized as a consequence of the workout. This phenomena could represent the psychological response to exercise, in which initial energy depletion is followed by a sensation of vitality and well-being. According to research, exercise increases the release of endorphins and other chemicals, which improves mood and energy levels. Regular physical activity also improves sleep quality, which increases energy and reduces fatigue (Harvard Health Publishing, 2020). Such a rebound effect may help explain the limited effects of fatigue on physical parameters like HR.

The study design also revealed limitations that influenced the results. A significant concern was the variability in timing for post-exercise assessments; as 38% of participants completed their measures more than 15 minutes after their workout. Prior studies indicate that the immediate effects of exercise can dissipate within this period of time, potentially obscuring that transient impacts of fatigue on gait metrics, as noted by Nardone et al. (2021). Furthermore, the self-guided nature of the exercise routine introduced heterogeneity in workout intensity and duration, which might reduce the reported effects of fatigue. Lander et al. (2009) found that self-paced exercise provides less physiological challenges than enforced constant pace exercise, implying that self-selected intensities may not provide consistent exhaustion levels across participants. Furthermore, several participants delayed the post workout assessments, diminishing data collecting consistency. Chau et al. (2005) emphasize that variability in gait data can result from errors in data gathering procedures, which may conceal effects. These findings

show that standardizing assessment time and training procedures is critical for effectively measuring the impact of fatigue on gait parameters.

It is also noted that the current study did not correct for family-wise alpha inflation. The consequence of not correcting for the multiple statistical tests is that some of the emergent findings may have emerged due to chance alone.

The implications of these results are significant for understanding fatigue in clinical and athletic contexts. However, these findings should be interpreted with consideration of the above noted limitations. The variability introduced by self-guided workouts and differences in post-exercise timing likely diminished the study's ability to detect stronger effects. Furthermore, the small sample size could have limited statistical power, especially in detecting small changes in the stability and symmetry domains. Relying simply on HR as a physiological measure limited the scope of fatigue assessment, as additional exertion markers, such as lactate levels or oxygen saturation, might provide a more complete picture of fatigue.

Future Directions

Future research should concentrate on a few important areas to expand on the findings of this study. First, studying the recovery process by assessing gait parameters at various time intervals post-exercise would provide useful information about the course of fatigue recovery and its impact on neuromotor performance. Expanding the sample size and to include participants with a variety of fitness levels, age groups, and physical conditions will improve the findings' generalizability and provide a better understanding of how individual differences influence fatigue and gait dynamics. Integrating additional physiological variables, such as lactate levels and oxygen consumption, could provide a more complete picture of the

mechanisms underlying fatigue and its consequences on gait. Longitudinal studies that follow people over time would also be useful, as they could indicate how recurrent exposure to fatigue affects gait adaptations and recovery mechanisms. Furthermore, studying the relationships between mental and physical fatigue may give insight into their combined impact on gait and neuromotor function. Such research would assist in determining whether the interaction of these two aspects of fatigue magnifies or modifies their individual effects. Finally, identifying fatigue thresholds that cause major impairments in gait stability, power, balance, and symmetry would be extremely useful in clinical and athletic applications, allowing for more accurate forecasts of performance limitations and injury risks across varied groups. Together, these directions would provide a more in-depth and complex knowledge of how fatigue affects human movement.

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